FACULTY SEMINAR · CALCULUS

Section 4.2 · Applications of Differentiation

Concavity and Inflection Points

$$f''(x) > 0 \Rightarrow \smile \qquad f''(x) < 0 \Rightarrow \frown$$

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Department of Mathematics

Agenda

01 Learning Objectives

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03 Interactive: Concavity Explorer

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06 Example 2

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08 The Second Derivative Test

09 Example 10 (1)

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11 Communicate with Mathematics 2

Learning Objectives

At the end of this section, you should be able to:

- Investigate additional features of graphs of functions, particularly concavity and points of inflection.

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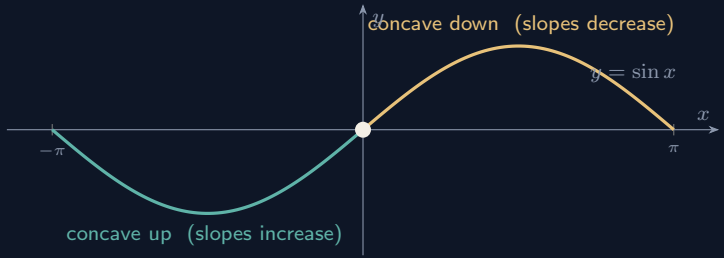
At the end of this section, you should be able to:

- Investigate additional features of graphs of functions, particularly concavity and points of inflection.
- Explore and analyze the concavity of graphs of functions using the methods of calculus.
- Identify and determine points of inflection of a function using calculus techniques.

Concavity

Interactive ↗

The first derivative is the slope of the tangent line; the **second derivative** describes how that slope changes — the *curvature* of the graph.



Concave up: the graph lies **above** its tangent lines.

Concave down: the graph lies **below** its tangent lines.

Definition 4.2.1 & Theorem 4.2.1

Definition 4.2.1 (Concave Up or Concave Down)

Let f be a differentiable function on an open interval I. Then

1. f is said to be **concave up** on I if f' is increasing on I.
2. f is said to be **concave down** on I if f' is decreasing on I.

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Theorem 4.2.1 (Concavity Test)

Suppose $f''(x)$ exists on an open interval I.

1. If $f''(x) > 0$ on I, then f is concave up on I.
2. If $f''(x) < 0$ on I, then f is concave down on I.

Interactive: Concavity Explorer

Drag a point along the curve and watch:

- the tangent line and its slope $f'(x)$,
- the sign of $f''(x)$,
- the curve coloured concave up / concave down,
- the inflection points marked.

Functions: the examples of this seminar.

Open the Concavity Explorer

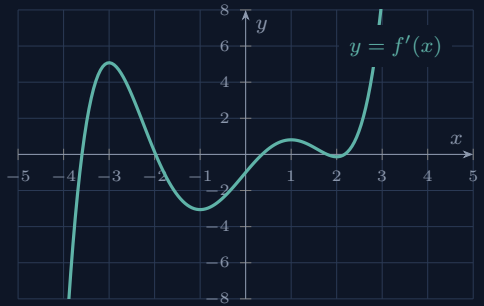
https://alostazy2k.github.io/seminars/4.2/concavity-explorer.html

offline copy (same folder as this PDF)

Related Problem 1

Interactive ↗

Use the graph of $y = f'(x)$, given in the figure, to find the following: 1. The intervals on which f is concave up. 2. The intervals on which f is concave down.



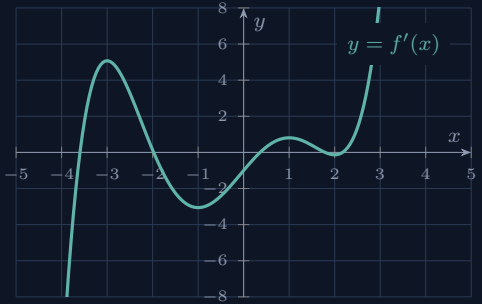
graph redrawn with the same turning points as Figure 4.2.4

Answer

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graph redrawn with the same turning points as Figure 4.2.4

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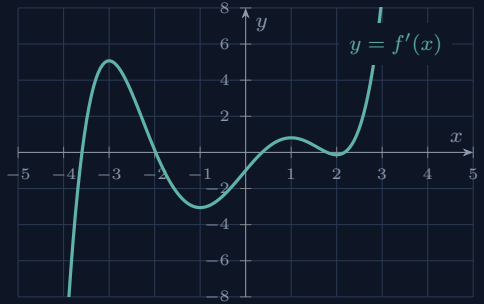
f' increasing on
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

$\Rightarrow$ **1. Concave up on**
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

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Answer

f' increasing on
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

$\Rightarrow$ **1. Concave up on**
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

f' decreasing on $(-3, -1)$, $(1, 2)$

$\Rightarrow$ **2. Concave down on** $(-3, -1)$, $(1, 2)$

Definition 4.2.2 & Theorem 4.2.2

Definition 4.2.2 (Inflection Point)

Let f be a continuous function on an open interval I and $x_0 \in I$. Then the point $(x_0, f(x_0))$ is said to be an **inflection point** if $f(x)$ changes its concavity (upward to downward or downward to upward) around x_0.

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Theorem 4.2.2 (Points of Inflection)

Let $(c, f(c))$ be a point of inflection of the graph of the function f. Then, either $f''(c) = 0$, or $f''(c)$ does not exist.

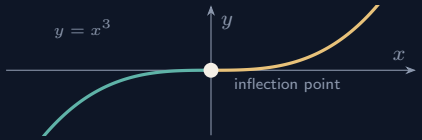
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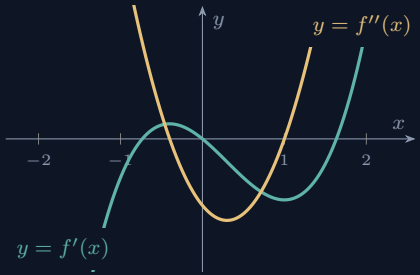
Let $(c, f(c))$ be a point of inflection of the graph of the function f. Then, either $f''(c) = 0$, or $f''(c)$ does not exist.



Example 2

Interactive ↗

Consider the graphs of $y = f'(x)$ and $y = f''(x)$. Determine: 1. the intervals on which f is concave up; 2. the intervals on which f is concave down; 3. the x-coordinates of all inflection points of $f(x)$ (if any).



graphs redrawn with the same zeros of f'' as Figure 4.2.7

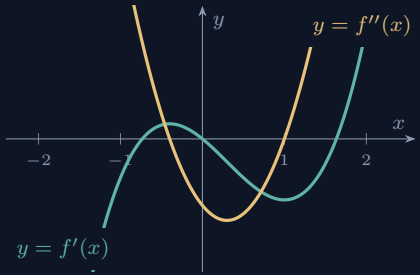
Solution

Using the graphs and the test for concavity:

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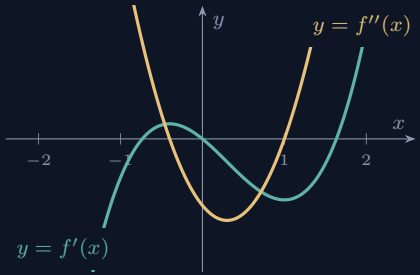
Using the graphs and the test for concavity:

1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$

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Solution

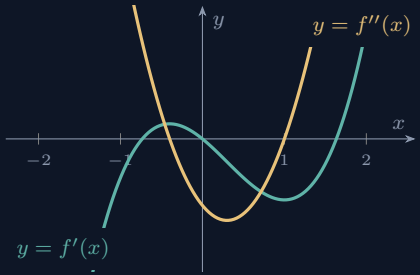
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1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$
2. f is concave down on $\left(-\frac{2}{5}, 1\right)$

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Solution

Using the graphs and the test for concavity:

1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$
2. f is concave down on $\left(-\frac{2}{5}, 1\right)$
3. Inflection points of f occur at

$$\boxed{x = -\tfrac{2}{5} \text{ and } x = 1}$$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

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Interval	$(-\infty, 2)$	$(2, \infty)$
Test value k	0	3
Sign of $f''(k)$	$-$	$+$
Behavior of $f(x)$	concave downward	concave upward

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$f(2) = 32 \implies$ $\boxed{(2,\ 32) \text{ is an inflection point}}$

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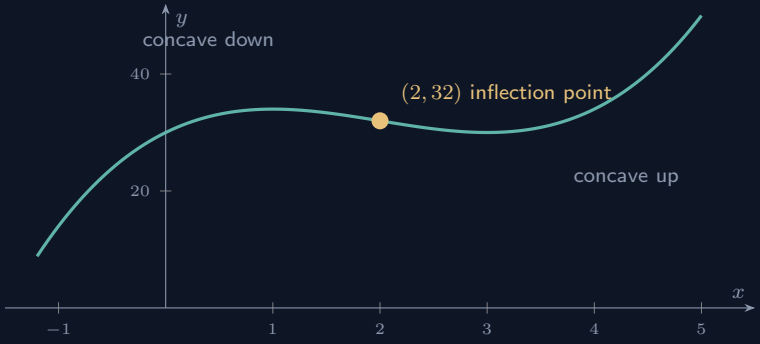
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$f(2) = 32 \implies$ $(2, 32)$ is an inflection point

Example 3 — graph

Interactive ↗



Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

Example 6

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x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

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Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

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No inflection points, although the concavity changes around $x = 5$,

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Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

No inflection points, although the concavity changes around $x = 5$,

because $5 \notin D_f$

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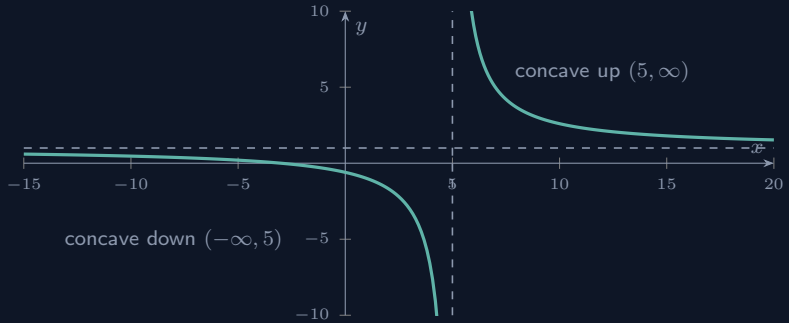
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Example 6 — graph

Interactive ↗



Example 8

Interactive ↗

For $f(x) = x^{2/3}(1 - x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

Example 8

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f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2 - 5x}{3x^{1/3}}$$

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$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

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x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

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$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0	$+\infty$
$f''(x)$		$+$		$-$		$-$
$f(x)$		$\smile$		$\frfrown$		$\frown$

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$-2(5x+1) = 0 \implies x = -\frac{1}{5} \implies f''\left(-\frac{1}{5}\right) = 0$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

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Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

f'' does not change sign around $x = 0$: no inflection at $x = 0$

f'' changes sign around $x = -\frac{1}{5}$ (positive to negative):

only inflection point = $\boxed{\left(-\frac{1}{5},\ \frac{6}{5}\sqrt[3]{\frac{1}{25}}\right)}$

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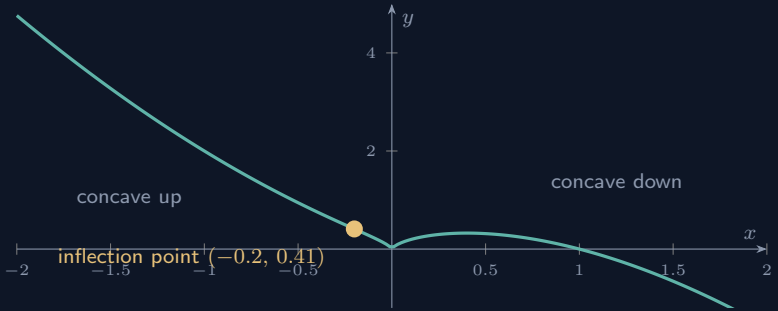
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Example 8 — graph

Interactive ↗



Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is continuous and twice differentiable on $[0, 2\pi]$

Example 9

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For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

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x	0		π		2π
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

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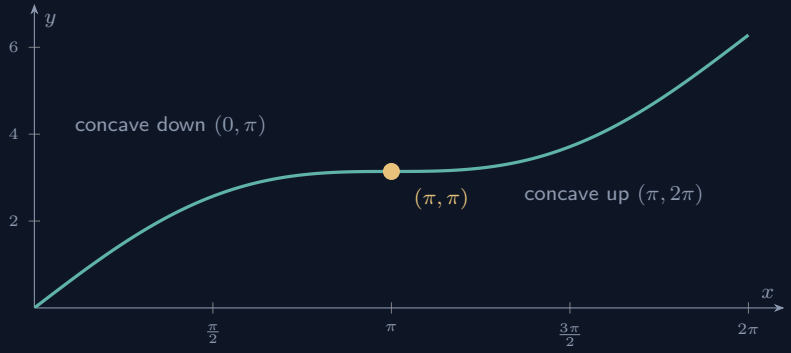
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Example 9 — graph

Interactive ↗



Theorem 4.2.3 (The Second Derivative Test)

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Suppose f'' is continuous on an open interval containing c with $f'(c) = 0$.

- If $f''(c) > 0$, then f has a relative minimum at c.
- If $f''(c) < 0$, then f has a relative maximum at c.
- If $f''(c) = 0$, then the test is inconclusive; f may have a relative maximum, relative minimum, or neither at c.

In such cases ($f''(c) = 0$), we may use the first derivative test.

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

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$$f'(0.5) = 4.5 > 0, \quad f'(1.5) = \tfrac{15}{2} > 0$$

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$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

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$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

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Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

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inflection points = $\boxed{\left(-\frac{1}{3},\ -\frac{94}{27}\right)}$, $\boxed{(1,\ 6)}$

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$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

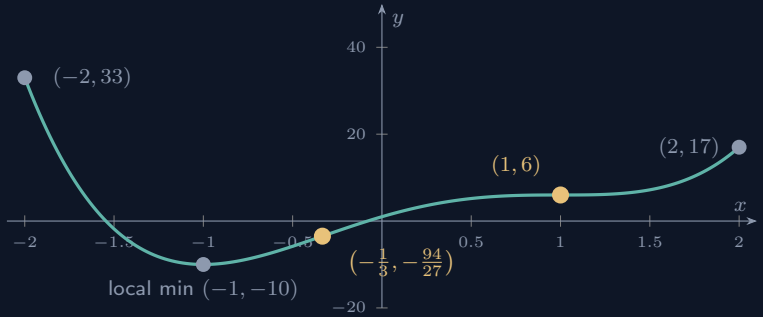
Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

Concavity changes at $x = -\frac{1}{3}$ and $x = 1$:

inflection points $= \boxed{\left(-\frac{1}{3},\ -\frac{94}{27}\right)},\ \boxed{(1,\ 6)}$

Example 10 (1) — graph

Interactive ↗



Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

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Solution f is quadratic $\implies a \neq 0$

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Thank you

Questions & discussion

Dr. Wessam Gouda · Section 4.2 — Concavity and Inflection Points